Contact

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Top Skills

Aerospace
Aeronautics
Aircraft

Tim W.

Lead Flight Test Engineer
Gatwick, England, United Kingdom

Summary

All views I express on LinkedIn are my own, and do not necessarily reflect those of my current employer or previous employers.

I am a free-thinking, creative, highly proficient and experienced professional aerospace engineer, technical manager, Lead Flight Test Engineer (EASA Cat 1 / Mil Class A) and flight test instructor, with aircraft certification experience, programme management experience and excellent people and communication skills.

I have worked in flight test since 2003. Highlights include: planning and leading deployed flight trials on Harrier GR7; being Lead Flight Test Engineer for the Pilatus PC-24 business jet programme, taking it from taxi tests and maiden flight through to initial type certification; being a senior instructor at ETPS; troubleshooting 2 high performance RPAS development programmes for QinetiQ; and working for the CAA.

I have 29 years' experience encompassing:

- aerospace engineering;

- test and evaluation of aircraft and RPAS/UAS, avionics and systems, including flight test planning, leadership and technical management.

- whole-aircraft flight test and certification (UK, EASA, FAA standards)

- systems safety analysis;

- flight simulation;

- lecturing / instructing / mentoring in aeronautical engineering and flight test, at degree and post-grad level.

I am an expert in air data systems and RVSM, aircraft performance, stability and control, flight control systems, autopilots, civil and military avionics, failure case testing, flight test instrumentation, and all associated test and certification activities, including flight test programme leadership, planning, conduct, management and safety.

I also have experience in simulation and modelling, and am familiar with both linear and non-linear flight models and control systems, using Matlab / Simulink.

Other professional interests include air vehicle design, BVLOS RPAS, urban and regional air mobility, and electrification/ hybridisation of aircraft propulsion.

I am a graduate of RAF College Cranwell and the Empire Test Pilots' School, with 200 hours as test conductor/telemetry room lead and 285 flight test engineer/observer flight hours on 30 aircraft types, ranging from light aircraft and helicopters through to military fast jets, business jets, large transport aircraft, and high performance RPAS. I also have a private pilot's licence.

I am a Member of the RAeS, the RAeS General Aviation Specialist Group committee, the Society of Flight Test Engineers, and Working Group 2 of the ICAO RPAS Panel (C2 Link).

Experience

Civil Aviation Authority

Design and Certification Specialist

January 2022 - Present (4 years 6 months)

Gatwick, England, United Kingdom

Working as a D&C specialist, Flight Test Engineer and Subject Matter Expert, primarily on Panel 1 (flight) and Panel 6 (avionics) for certification and validation projects concerning UK registered aircraft and new initial type certification projects. Focus is mostly on aircraft certified under CS-25 and equivalent international standards, including CS-ACNS and CS-AWO. Also the CAA technical and airworthiness lead for Certified Category RPAS, and

member of the UK delegation to the ICAO RPAS Panel, focusing particularly on C2 Link, Required Link Performance, and Detect and Avoid.

CAA deputy lead for C2 Links, C2 Communications Service Provision, HAPS / Higher Airspace Operations. Technical advisor to several elements of the UK Airspace Modernisation and Future of Flight programmes, including Electronic Conspicuity, Future ATM/ANS, UTM, DAA, CNS and AAM, focusing on air vehicle capabilities, airspace integration and functional safety.

Also assisting with UK policy development for eVTOL / VCA design, certification and operations, All Weather Operations (AWO), and potential future improvements to aircraft TAWS and navigation systems.

Part 21 Specialist Auditor for Flight Conditions, Permit to Fly and Flight Test.

QinetiQ

4 years 6 months

Flight Test Engineer and Senior Fixed Wing Flight Dynamics Instructor, ETPS

January 2020 - December 2021 (2 years)

Lead flight dynamics instructor for fixed wing courses. Primary responsibilities were planning/managing the fixed wing ground school programme, teaching and mentoring students, assessing and reporting on their progress, and developing new course modules and exercises. Also was a flight simulator engineer/operator for ETPS, and occasionally took on the role of FTE instructor for some of the student training exercises.

Periodically worked as a Flight Test Engineer or flight physics specialist for the wider QinetiQ business, effectively providing internal consultancy. Projects of note included: RVSM analysis and certification advice for a large multi-engine fixed wing aircraft; advising on technical requirements for a future flight simulation project; assessment of the flare law on an A350 Full Flight Simulator; and provision of flight physics analysis and advice regarding the stability, control and performance of a variety of small, high-speed, fixed wing UAVs.

Flight Test Engineer, Simulator Engineer and Instructor

March 2019 - December 2021 (2 years 10 months)

Boscombe Down

Teaching air data systems, aerodynamics, flight mechanics, aircraft performance, stability and control and flight test to student test pilots and flight test engineers.

Flight Dynamics Tutor - Empire Test Pilots' School
July 2017 - January 2020 (2 years 7 months)
Boscombe Down, Salisbury

Pilatus Aircraft Ltd
Lead Flight Test Engineer
September 2014 - June 2017 (2 years 10 months)
Stans, Switzerland

FTE for various performance and avionics upgrades on PC-12NG, then from early 2015 became Lead FTE on the PC-24 programme. Planned and conducted the first flight and many of the subsequent early development flights on the first PC-24 prototype, including envelope opening, handling qualities, performance, flutter testing and systems development.

Subsequently worked on the second PC-24 prototype, planning and conducting flight director/autopilot/AFCS development and certification testing, and leading the systems integration testing, avionics and electrical failure case testing, initial 'hot and high' performance testing and air data system pressure error measurement/validation tests.

In 2017 took over responsibility for the third PC-24 prototype, planning and conducting the maiden flight and subsequent early envelope opening and systems flight test campaign, including RVSM certification flights, then resumed testing on the second prototype, leading the certification tests for External Noise in accordance with FAR/CS 36, and preparations for steep approach flight testing. Also planned, managed and flew several flight evaluations of the PC-24 prototypes with test crews from EASA and the FAA.

Farnborough College of Technology
Lecturer - Aeronautical Engineering
September 2011 - August 2014 (3 years)

Senior lecturer and examiner in aeronautical engineering. Taught BEng degree subjects including aircraft systems, aircraft performance, aircraft conceptual design, aerodynamics, electronics and microprocessor systems, and supervised final year projects and dissertations.

Also taught maths, theory of flight, gas turbine engines and microprocessor systems to aircraft engineering apprentices, employed by companies such as British Airways, TAG at Farnborough and 2Excel Engineering at Lasham.

QinetiQ
Flight Test Engineer
July 2003 - May 2010 (6 years 11 months)
Boscombe Down

Worked in a variety of technical specialist / technical leadership roles on large multi-agency projects with multi-million pound budgets, driving through technical decisions to ensure the best possible results for each project.

2008-2010: Lead Flight Test Engineer and avionics specialist for Heavy Aircraft Evaluation Services (HAES). Departmental technical expert on Precision Area Navigation (P-RNAV), Flight Management Systems and Reduced Vertical Separation Minimum (RVSM). Technical lead on various projects including: resolution of altimeter system errors affecting RVSM clearance on a fleet of tanker/transport aircraft for the RAF; test and evaluation of various large aircraft navigation systems against civil aviation certification requirements; and ageing aircraft safety assessments for the RAF VC-10 fleet. Occasional lecturer at the Empire Test Pilots' School (ETPS), and departmental mentor for junior trials officers and test observers.

2007: selected for and successfully graduated from the year-long Flight Test Engineer course at ETPS.

2003-2006: Avionics and mission systems trials officer on the Harrier GR7/ GR9 avionics upgrade programme. Worked on several major flight test programmes associated with mission systems software, hardware and weapons upgrades. Also worked on various mission system software and hardware safety evaluation projects. Specialist in EO/IR sensors, targeting, HMI and systems integration. Planned and led several complex multi-agency trials programmes, including deployed flight trials and live weapon firing trials.

Royal Air Force (RAF)
Engineering Officer
August 1995 - March 2003 (7 years 8 months)
HQ Air Command | MOD London | Various RAF Stations

A variety of roles including:

C2ISR Role Office at RAF High Wycombe. Highlights included: planning and leading the redeployment of an air defence radar as part of UK air defence restructuring after 9/11; leading audits and training in the Falkland Islands, and making significant contributions to improving the engineering infrastructure at RAF Fylingdales.

Briefing officer at MOD London, providing briefings and preparing presentations for senior officers up to 2* level.

Engineering Officer on a Tornado GR1 Squadron, responsible for leadership and management of 80 engineering technicians, and the first line engineering support for up to 20 aircraft, including airworthiness sign-off and flight safety responsibilities.

Education

University of Southampton

Bachelor of Engineering (BEng), Mechanical Engineering · (1992 - 1995)